MAKE-UP EXAM

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BPLCK205C/ BPLCKC205

Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023

Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain object oriented principles.	6	L2	CO1
	b.	Describe the meaning of each of the keyword in "public static void main"	6	L2	CO1
	c.	Explain different lexical issues in JAVA.	8	L2	CO1
		OR			
Q.2	a.	Explain different promotion rules in JAVA.	6	L2	CO2
	b.	Explain one dimensional and two dimensional array with example for each.	6	L2	CO2
	c.	Write short note on integer literals.	8	L2	CO2
		Module – 2			
Q.3	a.	Explain the following operators with example i) << ii) >> iii) >>> iv) &	8	L2	CO2
	b.	Demonstrate the working of for loop.	6	L2	CO2
	c.	Write a program to sort the elements using for loop.	6	L2	CO2
		OR			
Q.4	a.	Explain Left shift and Right shift bitwise logical operations.	8	L2	CO2
	b.	Explain the syntax of switch statement with example.	6	L2	CO2
	c.	Write a program to find biggest of three members using ternary operator.	6	L2	CO3
		Module – 3			
Q.5	a.	What are constructors? Explain two types of constructors.	8	L2	CO2
	b.	Explain "this" keyword by considering box( ) version.	6	L2	CO2
	c.	Write a JAVA program to perform stack operation using proper class and methods.	6	L2	CO3
		OR			
Q.6	a.	Demonstrate the working of Recursion with example.	10	L2	CO3
	b.	Write a JAVA program demonstrating method overloading.	10	L2	CO3

		Module – 4			
Q.7	a.	Explain different types of inheritance in JAVA.	8	L2	C
	b.	Describe the uses of super keyword with example.	6	L2	C
	c.	Demonstrate the working of final class and final method in JAVA.	6	L2	C
		OR			
Q.8	a.	What is abstract class? Explain with example.	8	L2	C
	b.	Give example for method overriding.	6	L2	C
	c.	Explain the method of object class with their purpose in Java.	6	L2	C
		Module – 5			
Q.9	a.	What is a package? Explain with example.	8	L2	C
	b.	Demonstrate the working of interface in JAVA.	6	L2	C
	c.	Explain the concept of access protection in JAVA.	6	L2	C
		OR			
Q.10	a.	What is an exception? Give an example.	6	L2	C
	b.	Explain the try and catch block with example.	6	L2	C
	c.	Explain Nested throw statement in JAVA.	8	L2	C

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BPLCK105C/BPLCKC105

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the basic principles of object oriented programming.	6	L1	CO1
	b.	Explain primitive data types used in java.	8	L1	CO1
	c.	What is type conversion and type casting? Explain with an example.	6	L1	CO1
		OR			
Q.2	a.	Explain two control statements used in Java with syntax and example.	6	L1	CO1
	b.	What is multidimensional array? Explain with an example program.	8	L1	CO1
	c.	Explain lexical issues used in java.	6	L1	CO1
		Module – 2			
Q.3	a.	Write a java program to show the use of the following operators: i) << ii) >> iii) >>>	6	L3	CO2
	b.	Explain logical operators with example program.	8	L1	CO2
	c.	Explain for each version of for loop with an example.	6	L3	CO2
		OR			
Q.4	a.	Write a note on: i) Break ii) Continue.	6	L1	CO2
	b.	Explain different types of if statement with example program.	8	L3	CO2
	c.	Write a program to demonstrate the use of ternary operator.	6	L3	CO2
		Module – 3			
Q.5	a.	Explain two ways of argument passing methods with example.	6	L1	CO3
	b.	What are constructors? Explain two different types of constructors with example program.	8	L3	CO3
	c.	Define method overloading. Explain constructor overloading with an example.	6	L2	CO3

		OR			
Q.6	a.	Write a program to perform stack operations using proper class and methods.	6	L3	C
	b.	What is recursion? Explain with an example program.	8	L1	C
	c.	Explain: i) This keyword ii) The finalize( )	6	L1	C
		Module – 4			
Q.7	a.	What is Inheritance? Explain how a super class variable can reference a subclass variable.	6	L2	C
	b.	Explain method overriding with example program.	8	L3	C
	c.	Describe the two uses of super keyword with example.	6	L3	C
		OR			
Q.8	a.	Write a note on: i) Final class ii) Final method.	6	L1	C
	b.	Explain dynamic method dispatch with example.	8	L3	C
	c.	What is abstract class? Explain with an example program.	6	L3	C
		Module – 5			
Q.9	a.	Define package. Explain with an example program.	6	L3	C
	b.	What is an exception? Explain different mechanisms of handling the exceptions in java.	8	L3	C
	c.	Explain nested interfaces with example.	6	L1	C
		OR			
Q.10	a.	Describe different access specifiers used in java.	6	L1	C
	b.	Define Interface. Demonstrate the working of interface.	8	L1	C
	c.	Explain how one interface can be extended by another with an example program.	6	L3	C

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BPLCK105C/BPLCKC105

First Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain any three object Oriental principles.	6	L2	CO1
	b.	Describe the process of compiling and running the Java application, with example.	6	L1	CO1
	c.	Discuss the various primitive data types used in Java.	8	L1	CO1
		OR			
Q.2	a.	Explain type conversion in Java with example.	6	L2	CO1
	b.	What are keywords and identifies? List the rules to write identifiers.	6	L1	CO1
	c.	How are array declared and initialized in Java? Explain with suitable examples.	8	L2	CO1
		Module – 2			
Q.3	a.	What are different types of operators in Java? Explain them.	6	L2	CO1
	b.	Write a program to find biggest among 3 numbers using ternary operator.	6	L1	CO1
	c.	Demonstrate working of break with labels in Java.	8	L2	CO1
		OR			
Q.4	a.	Differentiate between while loop and do While loop in Java.	6	L2	CO1
	b.	With program, explain continue and return keywords in Java.	6	L2	CO1
	c.	Write a program to calculate the average among the elements 4, 8, 10, 12 using for each in JAVA.	8	L3	CO1
		Module – 3			
Q.5	a.	Distinguish between Method Overloading and Method Overriding in Java, with suitable example.	8	L2	CO3
	b.	Mention and explain the use of This keyword in Java.	6	L2	CO1
	c.	What are Constructors? Explain two types of constructors.	6	L2	CO1

Q.6	a.	Explain Garbage collection in Java.
	b.	Explain how to perform stack operation using proper class and methods.
	c.	Mention and explain the use of final keyword in Java.

Module – 4

Q.7	a.	What is Inheritance? Discuss different types of inheritance with suitable example.
	b.	Explain Abstract class and Abstract method with an example.
	c.	Demonstrate the working of final class and final method in Java.

OR

Q.8	a.	Explain two uses of super keyword with example.
	b.	Demonstrate dynamic dispatch with abstract class in Java.
	c.	Explain Object class in Java.

Module – 5

Q.9	a.	Explain the steps involved in creating user defined package with an example.
	b.	Explain the syntax of throw and throws with example.
	c.	How do you define variables inside interface? List out the characteristics of such variables.

OR

Q.10	a.	What is an Interface? Explain how to define and implement interface by taking suitable example.
	b.	How do you create your own exception class? Explain with program.
	c.	Explain the use of try and catch block in JAVA.

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BPLCK105C/ BPLCK

8/13

First Semester B.E./B.Tech. Degree Examination, June/July 2024
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Describe the following control flow statements with sample program: (i) If (ii) for	08	L1	CO1
	b.	List the primitive types of data and describe any two of the data types with example.	08	L1	CO1
	c.	Define Type conversion and casting in Java.	04	L1	CO1
		OR			
Q.2	a.	Define Array and its types. Write a Java program for multiplication of two arrays.	10	L1	CO1
	b.	Explain the following terms: (i) Whitespace (ii) Identifiers (iii) Comments (iv) Separators with its purpose.	06	L2	CO1
	c.	Describe the term variables in Java programming along with its declaration and initialization.	04	L1	CO1
		Module – 2			
Q.3	a.	Summarize the Bitwise operators and demonstrate the following operations and sign extension with Java programs: (i) << (ii) >> (iii) >>>	10	L2	CO2
	b.	Summarize the jump statements supported by Java with example.	10	L2	CO2
		OR			
Q.4	a.	List and define all iteration statements in Java programming. Write a Java program to sort list of elements in ascending and descending order using for loop.	10	L2	CO2
	b.	Demonstrate the following operators with Java program: (i) Basic arithmetic operator (ii) Modulus operator (iii) Arithmetic compound assignment operator.	10	L2	CO2
		Module – 3			
Q.5	a.	Define the following with syntax: (i) Simple class (ii) Declaring objects (iii) Introducing methods Create a Java class called student with the following details as variables within it USN, Name, Branch, Phone, Percentage. Write a Java program to create n student objects and print the objects with suitable headings.	10	L1	CO3
	b.	Explain constructor and parameterized constructor with example.	10	L2	CO3
		OR			
Q.6	a.	Compare Method Overloading with constructor overloading. Demonstrate the method overloading and the constructor overloading using Java program.	10	L2	CO3
	b.	Explain the concept of recursion with example as computing the factorial of a number.	10	L2	CO3

	Module – 4			
a.	Design a super class called staff with details as StaffId, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (skills) and Contract (period). Write a Java program to read and display at least 3 staff objects of all three categories.	**10**	**L3**	
b.	Describe the use of 'final' keyword with inheritance in Java programming with a sample program.	**10**	**L1**	
	OR			
a.	Demonstrate dynamic dispatch using abstract class in Java program with example.	**10**	**L2**	
b.	Summarize the concept, method overriding with example program in Java.	**10**	**L2**	
	Module – 5			
a.	Explain the concept of package and write a Java program to create two packages P_1 and P_2. In package P_1, create class A, class B inherited from A, class C. In package P_2, create class D inherited from class A in package P_1 and class E. Demonstrate the working of access modifiers (private, public, protected, default) in all these classes using Java.	**10**	**L2**	
b.	Describe chained exceptions in Java programming.	**06**	**L1**	
c.	Paraphrase the concept of importing packages in Java programming.	**04**	**L2**	
	OR			
a.	Write a Java program to read two integers a and b. Compute a/b and print, when b is not zero. Raise an exception when b is equal to zero. Also demonstrate the working of Array Index Out Of Bound – Exception in Java programming.	**08**	**L2**	
b.	Explain the concept of interfaces with its implementation using Java program.	**08**	**L2**	
c.	State whether the try statement can be nested. If yes, give an example program. If no, infer the statement.	**04**	**L2**	

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BPLCK105C

First Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025
Basic of Java Programming

Time: 3 hrs. Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
1	a.	Discuss the different primitive data types available in Java. Provide examples for each.	6	L1	CO1
	b.	Differentiate between stack memory and heap memory in Java.	4	L1	CO1
	c.	Define arrays in Java. Write a program to initialize an array and find the sum of all elements.	10	L2	CO1
		OR			
2	a.	Explain the key features of Java as an object – oriented programming language.	4	L1	CO1
	b.	Explain different lexical issues in Java.	8	L1	CO1
	c.	Write a Java program that prints all real solution to the quadratic equation : $ax^2 + bx + c = 0$ Read in a, b, c and use the quadratic formula.	8	L2	CO1
		Module – 2			
3	a.	Explain the significance of the ternary operator in Java. Write a code snippet to demonstrate its usage.	6	L1	CO2
	b.	Explain the different types of control statement in Java and provide examples.	6	L1	CO2
	c.	Write a Java program to compute the factorial of a number using loop.	8	L2	CO2
		OR			
4	a.	Write a program to sort the elements using for loop.	10	L2	CO2
	b.	Write a Java program for multiplication of two arrays.	10	L2	CO2
		Module – 3			
5	a.	What are constructors? Explain two types of constructors.	6	L1	CO1
	b.	Explain use of this in Java.	6	L1	CO1
	c.	Write a Java program demonstrating method overloading.	8	L2	CO1
		OR			
6	a.	Explain static variable and static methods in Java.	6	L1	CO1
	b.	Explain memory allocation and use of garbage collector in Java.	6	L1	CO1
	c.	Write a Java program that creates a class Student with attribute name, rollno, and marks and displays the details of Student.	8	L2	CO1
		Module – 4			
7	a.	Define inheritance and its types in Java. How does Java support single and multilevel inheritance.	10	L1	CO3
	b.	Explain the concept of method overriding with a Java program.	10	L1	CO3

		OR			
8	a.	Demonstrate the working of final class and final method in Java.	**8**	**L1**	**CO3**
	b.	Explain two uses of super keyword with example.	**6**	**L1**	**CO3**
	c.	Write a Java program demonstrating constructor overloading.	**6**	**L2**	**CO3**
		Module – 5			
9	a.	What is a package? Explain with example.	**6**	**L1**	**CO4**
	b.	Explain different access modifiers in Java.	**6**	**L1**	**CO4**
	c.	Explain the try, catch and finally blocks in Java. Provide code to demonstrate.	**8**	**L2**	**CO4**
		OR			
10	a.	What is an interface in Java? Explain how interfaces are different from abstract classes.	**8**	**L1**	**CO4**
	b.	Write a Java program to read two integers a and b. compute a/b and print, when b is not zero. Raise an exception when b is equal to zero. Also demonstrate its working of ArrayIndexOutofBoundException.	**12**	**L2**	**CO4**

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BPLCK205C

12/13

Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Write a Java Program to find greatest of 3 numbers.	6	L2	CO1
	b.	With reason identify the following declarations are valid or invalid : #AvgTemp, Count, a4, Stest, 2Count, high-temp, _Sum , if	6	L2	CO1
	c.	Explain type casting and type conversion in Java with an example program.	8	L2	CO1
		OR			
Q.2	a.	Write a Java Program to swap data in 2 variables without using temporary variable.	4	L2	CO1
	b.	Define Array. With Syntax and Example Program. Explain declaration of single dimensional array in Java.	6	L2	CO1
	c.	Briefly explain the basic principle of Java.	10	L2	CO1
		Module – 2			
Q.3	a.	Briefly explain the following operators in Java with an example program : >> << ~	6	L2	CO2
	b.	With example program, explain following statements : (i) Break (ii) Continue	8	L2	CO2
	c.	Write a Java Program to find largest of 2 numbers using ternary operator.	6	L3	CO2
		OR			
Q.4	a.	Write a Java Program that prints all real solutions to the quadratic equation, $ax^2 + bx + c = 0$. Read in a, b, c and use the quadratic formula.	10	L3	CO1
	b.	With syntax and example program, explain iterative statements in Java.	10	L2	CO1
		Module – 3			
Q.5	a.	Explain following keywords in Java with examples : (i) this (ii) final (iii) Static	10	L3	CO3
	b.	Define Constructor. Explain types of constructors with example programs.	10	L3	CO3

		OR			
Q.6	**a.**	Explain Garbage collection in Java with an example.	**6**	**L2**	**CO3**
	b.	Write a Java program to find factorial of a number using recursion.	**6**	**L3**	**CO3**
	c.	Define method overloading. Explain with an example program.	**8**	**L3**	**CO3**
		Module – 4			
Q.7	**a.**	Why multiple inheritance is not supported in Java. Justify with an example.	**8**	**L3**	**CO3**
	b.	Explain the "Super" keyword in Java.	**4**	**L2**	**CO3**
	c.	Define method overriding. Explain with an example program.	**8**	**L2**	**CO3**
		OR			
Q.8	**a.**	Explain types of inheritance in Java with example.	**10**	**L2**	**CO3**
	b.	Design a Super Class called staff with details as staffId, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (Skills) and Contract (Period). Write a Java Program to read and display at least 3 staff objects of all three categories.	**10**	**L3**	**CO3**
		Module – 5			
Q.9	**a.**	With a neat block diagram, explain types of Java Exceptions.	**10**	**L2**	**CO4**
	b.	Define interface. Differentiate between Class and interface.	**5**	**L2**	**CO4**
	c.	Write a note on packages in Java.	**5**	**L2**	**CO4**
		OR			
Q.10	**a.**	Explain following types of Exceptions : (i) ArithmeticException (ii) ArrayIndexOutOf BoundsException (iii) SQLExceptions (iv) NullPointerExceptions	**12**	**L3**	**CO4**
	b.	Illustrate "throw" and "throws" in Java Exceptions with example.	**8**	**L3**	**CO4**

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